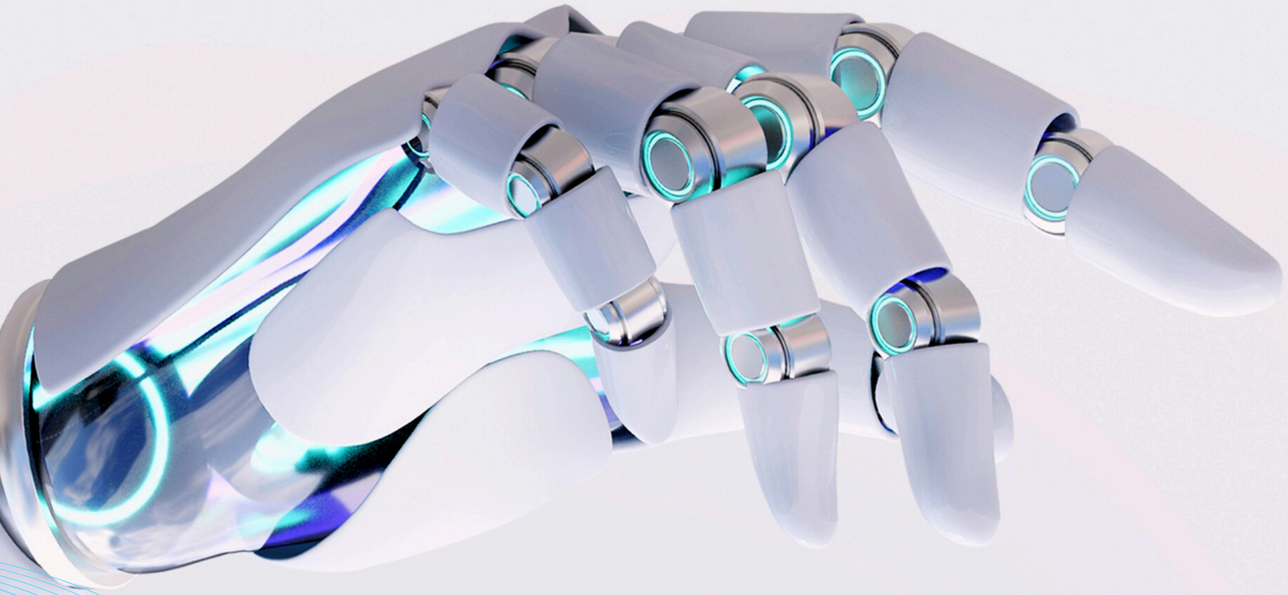




Deaf Reception Assistant

The Problem

Deaf patients often face communication challenges when interacting with hospital reception staff.
This communication gap may lead to:

- Delayed assistance
- Misunderstood medical needs
- Less efficient healthcare services

Our goal is to provide a simple AI-powered solution that translates medical sign language gestures into understandable medical terms.



Project Objective

To develop a Computer Vision system capable of recognizing Arabic medical sign language gestures and converting them into meaningful medical terms to assist hospital reception staff in communicating with deaf patients.

Expected Impact

- Improve communication accessibility
- Reduce misunderstandings
- Enhance patient experience
- Support healthcare inclusivity

Dataset

Dataset: KArSL-100 (RGB)
We selected 16 medical sign classes related to healthcare services.

Dataset Statistics

- Dataset Type: RGB
- Selected Classes: 16
- Total Images: 33,701
- Total Samples: 672
- Balanced Dataset



	الكلمة	عدد الصور
0	معاشقني	1847
1	سريالية	1454
2	مريض	1651
3	دواء	1354
4	تحليل دم	2362
5	فحص سريري	2967
6	فحص النظر	2463
7	سماعة أذن	1690
8	صداع	2502
9	ألم	1630
10	حمى	1931
11	تطعيم	1388
12	عملية جراحية	2671
13	إعاقة سمعية	1947
14	عدوى	1921
15	صورة اشعة	3923
إجمالي الصور:		33701

Data Preparation

Medical Class Selection

• Extracted 16 healthcare-related sign classes from the original KArSL-100 dataset.

• Selected classes represent common hospital communication needs.

Dataset Splitting

The dataset was divided into:

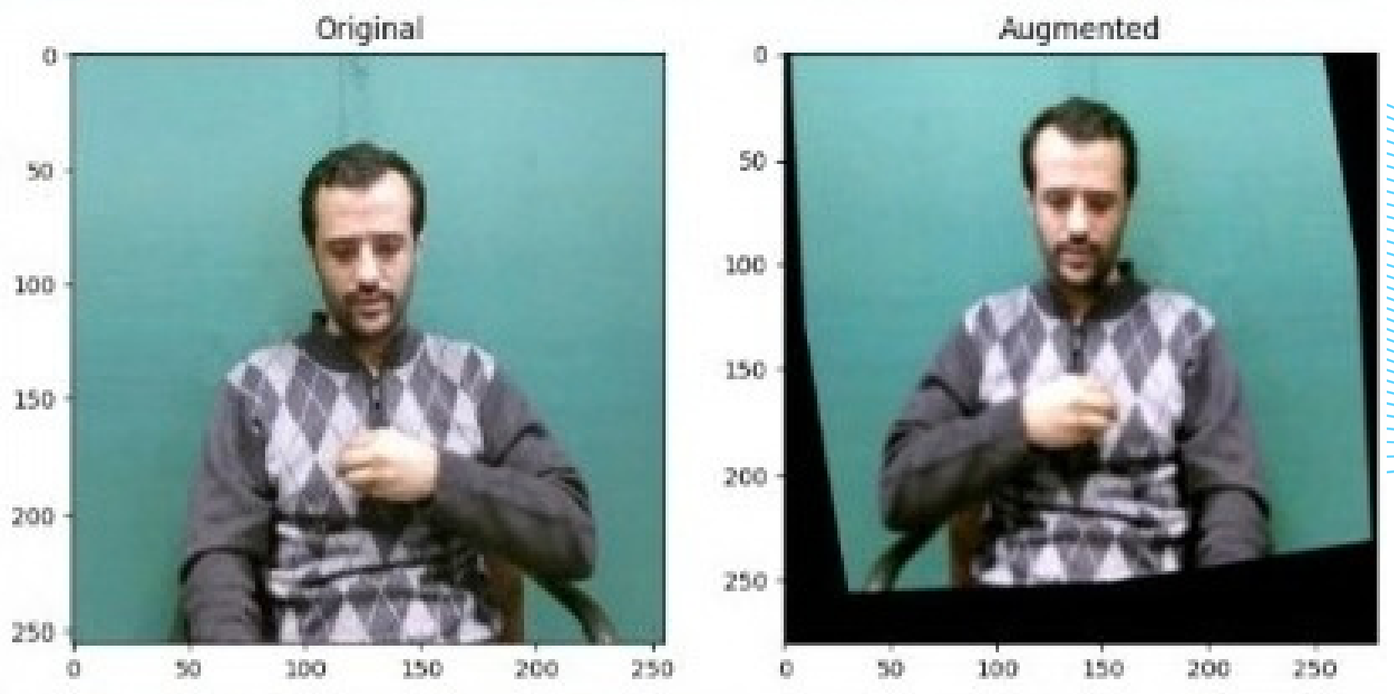
Dataset Splitting

- Training Set: **70%**
- Validation Set: **15%**
- Test Set: **15%**

Data Augmentation Techniques

To improve model generalization and reduce overfitting, six augmentation techniques were applied to the training set only:

1. Horizontal Flip
2. Rotation
3. Brightness Adjustment
4. Shift Transformation
5. Scaling
6. Gaussian Noise



MobileNetV2 with Transfer Learning

Why MobileNetV2?

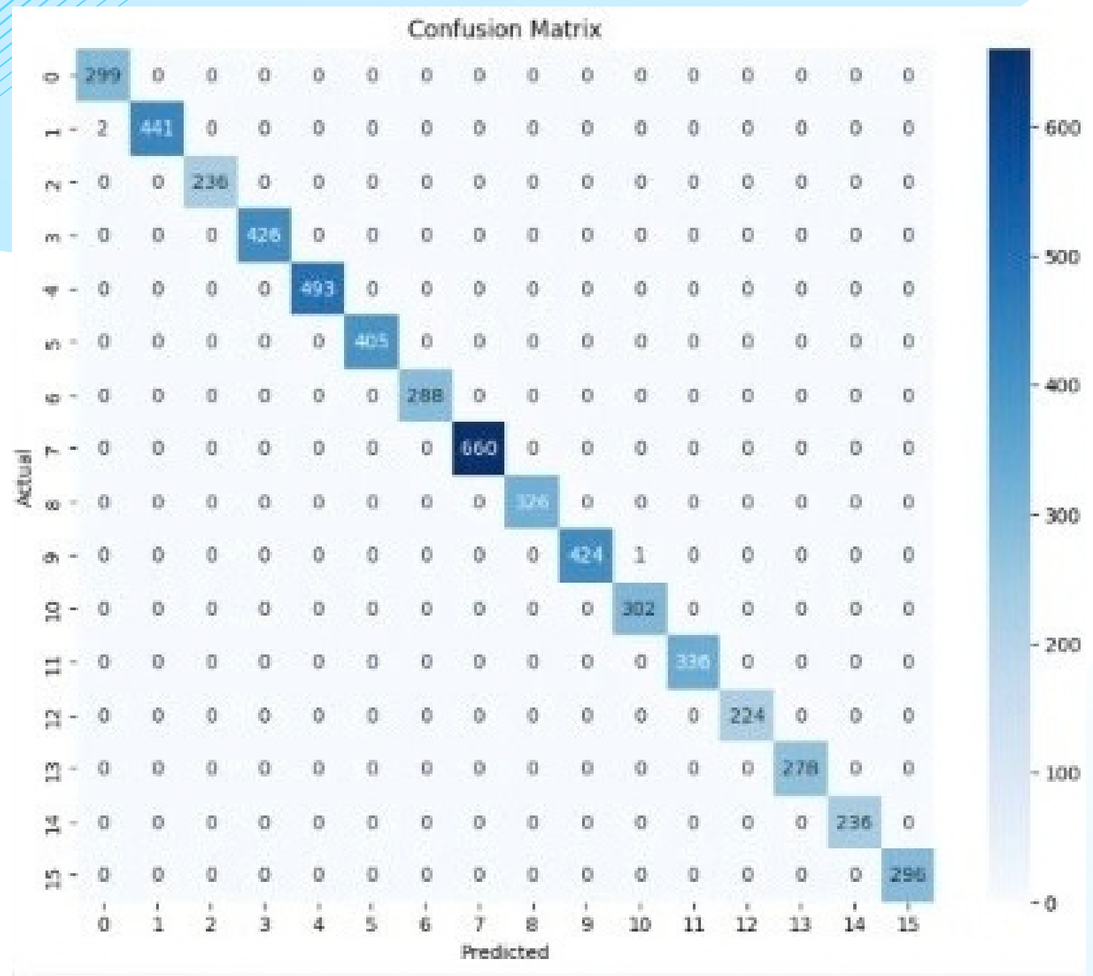
- Lightweight architecture
- Fast training
- High accuracy
- Suitable for real-time applications

Transfer Learning

Instead of training from scratch, we used a pre-trained MobileNetV2 model and fine-tuned it for our 16 medical sign classes.

Model Performance

confusion matrix



Training Performance

